Kepler's Three Laws: Shape, Speed, and Time

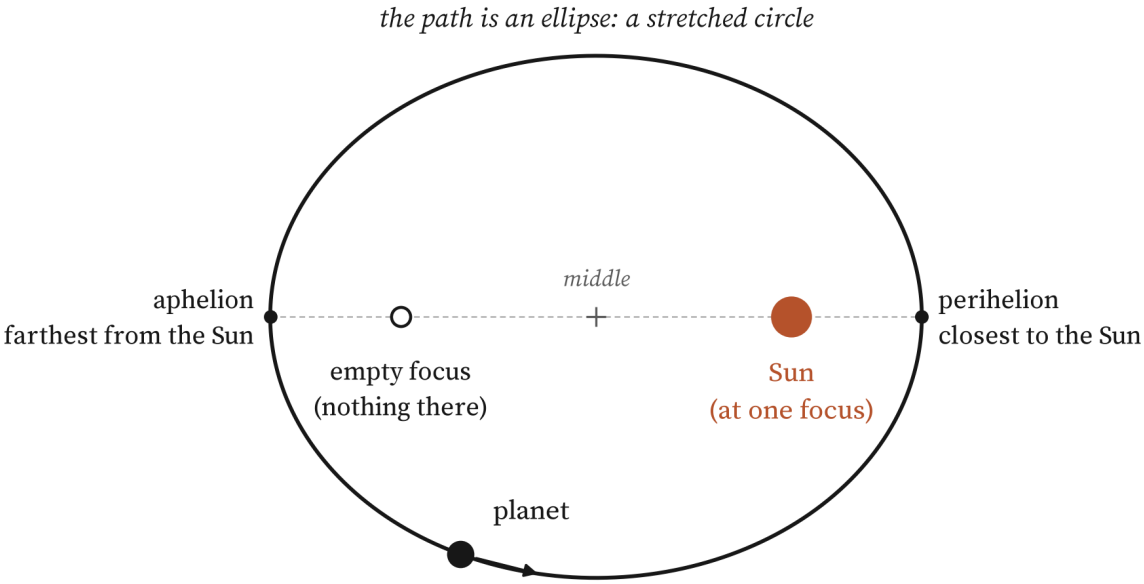
How the planets move around the Sun, in plain words

Johannes Kepler worked in the early 1600s. He used Tycho Brahe's careful observations, especially of Mars. He published the first two laws in 1609 and the third in 1619.

Law 1 · The Shape (1609)

A planet travels on an **ellipse**, not a perfect circle. An ellipse is a stretched circle, like an oval racetrack.

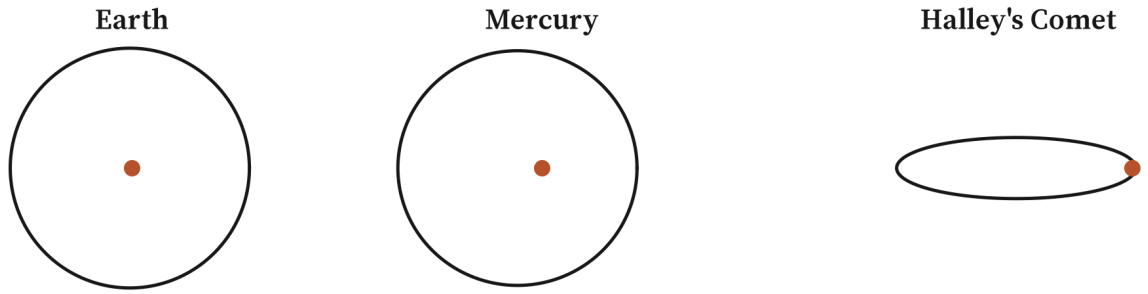
An ellipse has two special points inside it called **foci** (one is a **focus**). The Sun sits at one focus, a little off to one side. It is not in the middle. The other focus is empty.



This drawing is stretched on purpose so the idea is easy to see. **Perihelion** *is the point closest to the Sun;* **aphelion** *is the farthest.*

Careful: Real Planet Paths Are Only a Little Stretched

Astronomers measure the stretch with a number called the **eccentricity**, *e*. 0 is a perfect circle. Close to 1 is a long, skinny oval. Earth's *e* is only 0.017, so its path is almost a circle. Books that draw Earth on a skinny oval are wrong. Here are true shapes, each with the Sun at a focus. Even Mercury looks round; its off-center Sun gives it away.

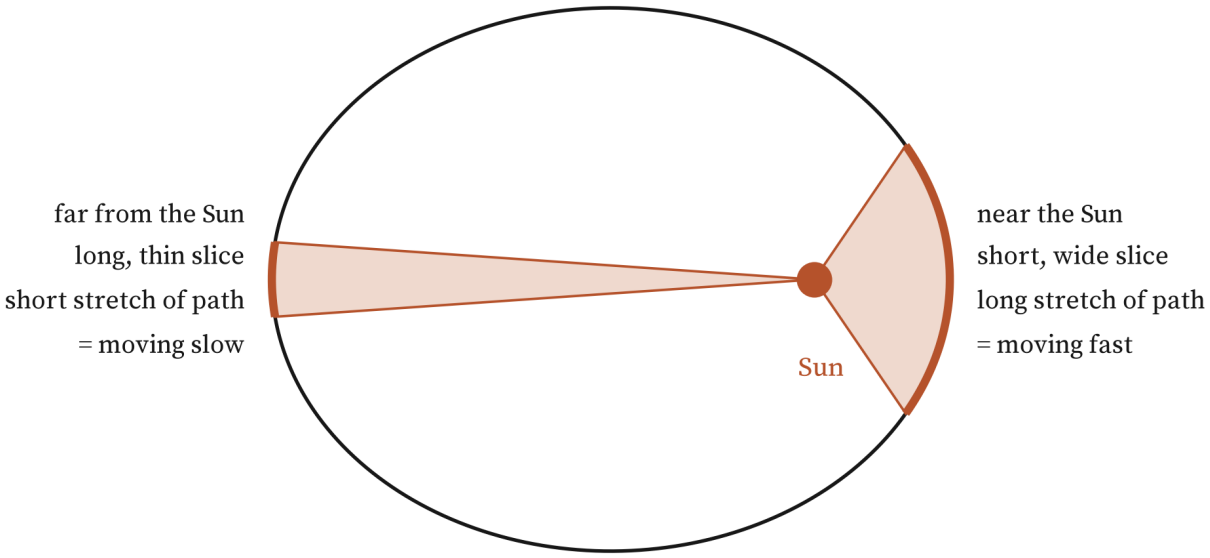


Law 2 · The Speed (1609)

Draw a line from the Sun to the planet. As the planet moves, that line sweeps out **equal areas in equal times**.

Think of the line as a windshield wiper. In one month it paints a slice of the ellipse. Every month's slice has the same area, the same amount of space inside it.

Both shaded slices: same time (about 26 days of a 1-year orbit), same area



Thick accent lines show how far the planet travels in that time.

When the planet is close to the Sun, it has to hurry. When it is far, it can take its time. The two slices have the same area, and they take the same amount of time.

Why the Slices Look So Different

Near the Sun the line is short. To cover the same area with a short line, the slice must be wide, so the planet covers a long stretch of path. Far from the Sun the line is long. A thin slice already holds the same area, so the planet covers only a short stretch.

Earth Does This Too

Earth is closest to the Sun in early January and farthest in early July. It moves a little faster in January than in July. That is one reason northern winter is a few days shorter than northern summer. (Seasons themselves come from Earth's tilt, not from distance.)

Earth's Speed Around the Sun	
Early January, closest (about 91.4 million miles)	about 67,800 mph
Early July, farthest (about 94.5 million miles)	about 65,500 mph

Earth's path is almost a circle, so its speed changes only a little: about 3 percent. A comet on a long, skinny path changes speed far more. It whips around the Sun fast, then crawls far out.

Law 3 · The Time (1619)

The farther a planet is from the Sun, the longer its year. In years and AU: $\mathbf{T^2 = a^3}$.

Words first. *T* is the planet's **period**: how long one trip around the Sun takes, in Earth years. *a* is its average distance from the Sun. (On the ellipse, *a* is half the long width, called the **semi-major axis**.) **1 AU**, one astronomical unit, is Earth's average distance from the Sun, about 93 million miles.

Square means a number times itself: $3^2 = 3 \times 3 = 9$. **Cube** means times itself, then times itself again: $3^3 = 3 \times 3 \times 3 = 27$.

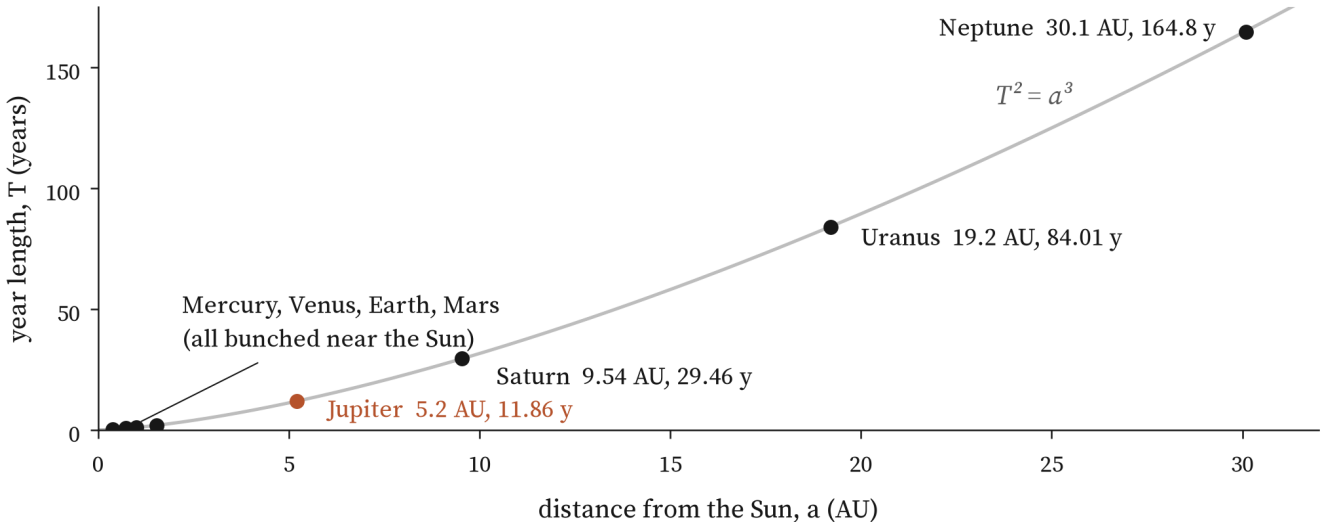
So the law says: the year times itself equals the distance times itself times itself.

Jupiter	**Mars** (a check)
Distance about 5.2 AU	Distance about 1.52 AU
$5.2^3 = 5.2 \times 5.2 \times 5.2 \approx 141$	$1.52^3 \approx 3.51$
$T^2 \approx 141$, so $T \approx \sqrt{141} \approx$ **11.86 years**	$T \approx \sqrt{3.51} \approx$ **1.87 years**; actual 1.88 (rounding)

√ means square root: the number that, times itself, gives what is under the sign. √9 = 3.

Planet	distance a (AU)	year T	a^3	T^2
Mercury	0.387	0.24 y (88 days)	0.058	0.058
Venus	0.723	0.62 y (225 days)	0.378	0.384
Earth	1	1 y (365 days)	1.00	1.00
Mars	1.524	1.88 y	3.54	3.53
Jupiter	5.2	11.86 y	141	141
Saturn	9.54	29.46 y	868	868
Uranus	19.2	84.01 y	7,078	7,058
Neptune	30.1	164.8 y	27,271	27,159

Look across each row: a^3 and T^2 come out nearly the same. The small gaps are rounding.



Why the Year Grows Faster Than the Distance

A farther planet has a longer path to travel, and it also moves more slowly along it. Both make the trip take longer. So twice as far is about **2.8 times** as long a year, not twice as long.